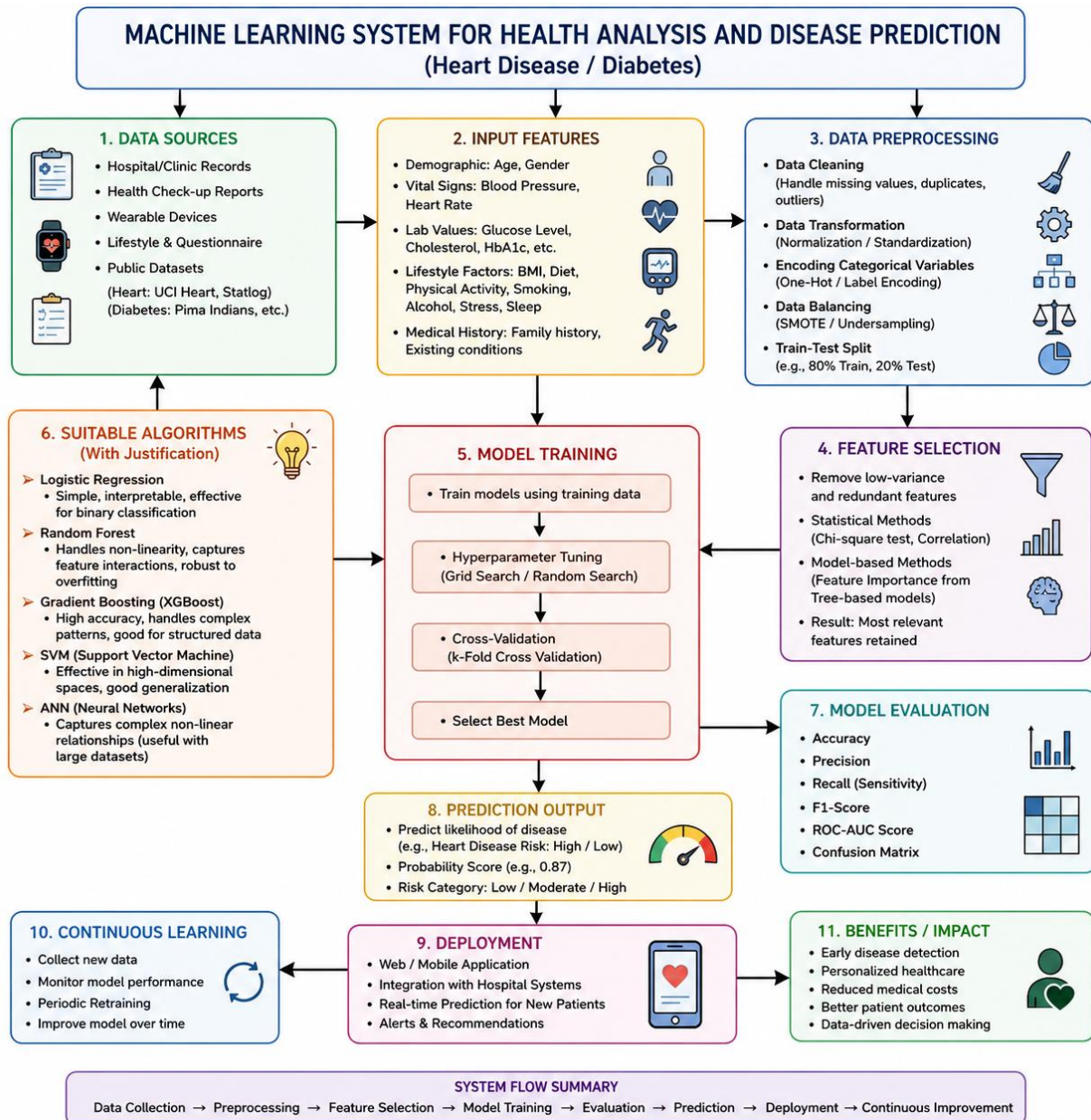


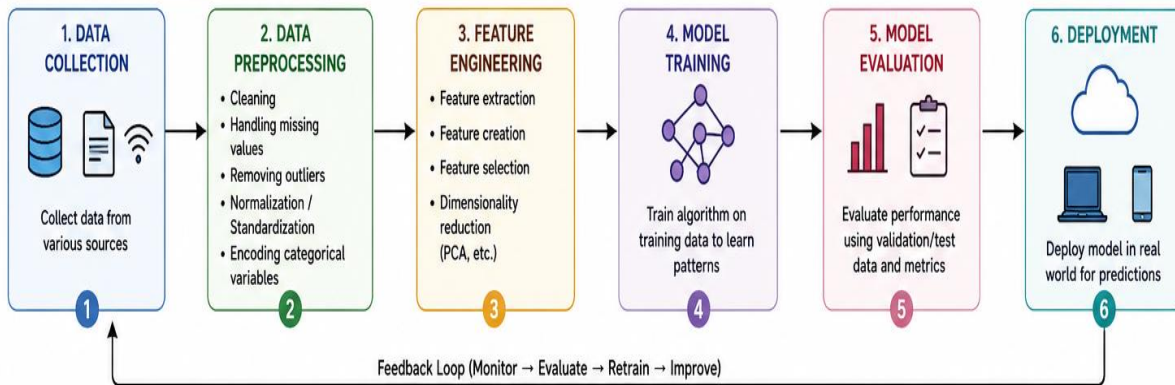
10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions. I want concept map



11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.

MACHINE LEARNING PIPELINE: ARCHITECTURE, STAGES & ERROR ANALYSIS

1. ARCHITECTURE OF A TYPICAL ML PIPELINE



2. ROLE OF EACH STAGE

	1. Data Collection Gather relevant, diverse and high-quality data from databases, sensors, APIs, surveys, logs, etc. The quality of data directly impacts model performance.
	2. Data Preprocessing Prepare raw data to make it suitable for modeling. Removes noise, handles inconsistencies, converts data to a proper format and scales features to improve model accuracy and stability.
	3. Feature Engineering Create new features or transform existing ones that better represent underlying patterns. Helps in improving model performance and reducing complexity.
	4. Model Training Use training data to learn the relationship between input features and target variable by optimizing model parameters using algorithms.
	5. Model Evaluation Assess how well the model performs on unseen data using metrics like Accuracy, Precision, Recall, F1-score, RMSE, AUC, etc. Helps in model selection and tuning.
	6. Deployment Integrate the model into applications or systems to make predictions on new data. Monitor performance in production and update model as needed.

3. TRAINING ERROR vs GENERALIZATION ERROR

Training Error (In-sample Error)

- Error measured on the training data.
- Indicates how well the model fits the training data it has seen.
- Always decreases (or stays same) as model complexity increases.
- Too low training error can indicate overfitting.

$$\text{Formula: } E_{\text{train}} = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i)$$

Example:

A decision tree with very deep branches may predict training data almost perfectly ($E_{\text{train}} \approx 0$) but perform poorly on new data.

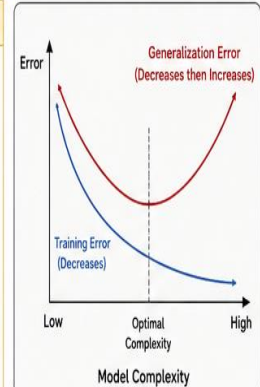
Generalization Error (Out-of-sample Error)

- Error measured on new, unseen data (validation/test set).
- Indicates how well the model performs on real-world data.
- Initially decreases with model complexity but after a point starts increasing (overfitting).
- Lower generalization error means better real-world performance.

$$\text{Formula: } E_{\text{gen}} = \frac{1}{m} \sum_{j=1}^m L(y_j, \hat{y}_j)$$

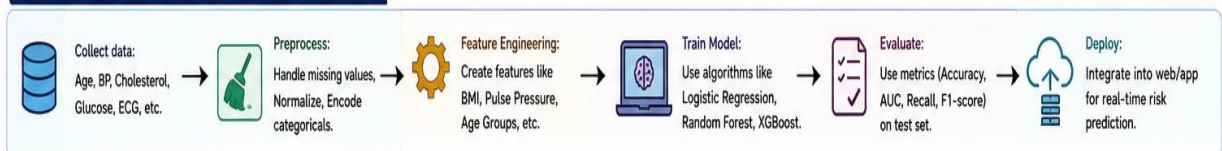
Example:

A model that memorizes training data will have high E_{gen} because it fails to generalize to unseen patients/customers.



Goal: Find the sweet spot where generalization error is minimum → best trade-off between underfitting and overfitting.

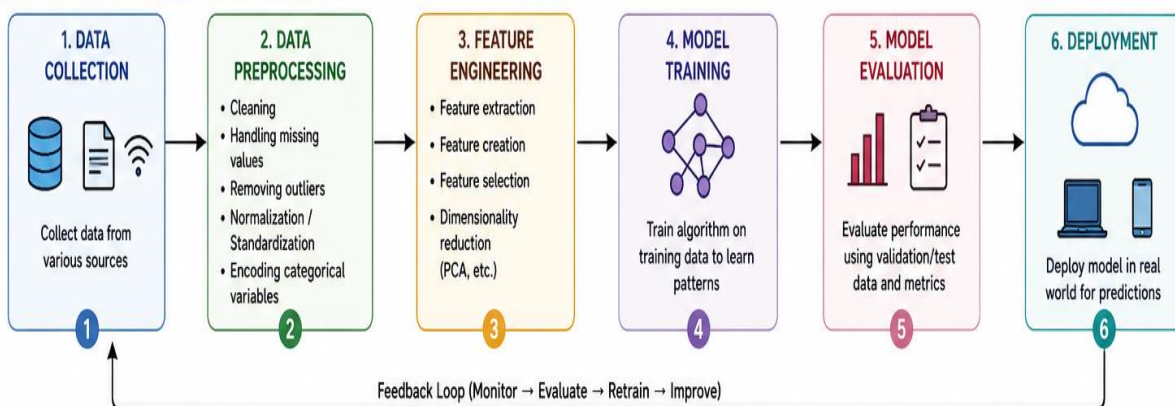
4. EXAMPLE OF PIPELINE (HEART DISEASE PREDICTION)



12. Design a machine learning model for spam email classification using labeled data. (5 Marks)(a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).(b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.(c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

MACHINE LEARNING PIPELINE: ARCHITECTURE, STAGES & ERROR ANALYSIS

1. ARCHITECTURE OF A TYPICAL ML PIPELINE

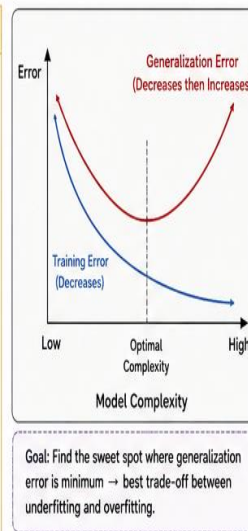


2. ROLE OF EACH STAGE

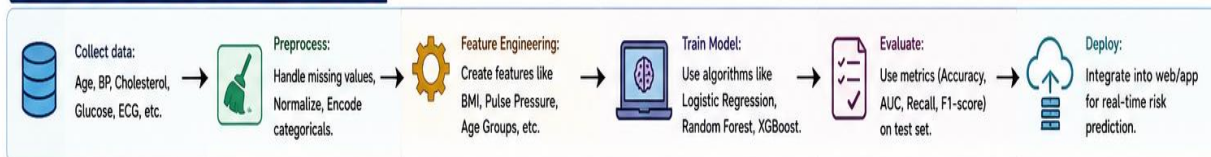
	1. Data Collection	Gather relevant, diverse and high-quality data from databases, sensors, APIs, surveys, logs, etc. The quality of data directly impacts model performance.
	2. Data Preprocessing	Prepare raw data to make it suitable for modeling. Removes noise, handles inconsistencies, converts data to a proper format and scales features to improve model accuracy and stability.
	3. Feature Engineering	Create new features or transform existing ones that better represent underlying patterns. Helps in improving model performance and reducing complexity.
	4. Model Training	Use training data to learn the relationship between input features and target variable by optimizing model parameters using algorithms.
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	6. Deployment	Integrate the model into applications or systems to make predictions on new data. Monitor performance in production and update model as needed.

3. TRAINING ERROR vs GENERALIZATION ERROR

Training Error (In-sample Error)	Generalization Error (Out-of-sample Error)
- Error measured on the training data. - Indicates how well the model fits the training data it has seen. - Always decreases (or stays same) as model complexity increases. - Too low training error can indicate overfitting.	- Error measured on new, unseen data (validation/test set). - Indicates how well the model performs on real-world data. - Initially decreases with model complexity but after a point starts increasing (overfitting). - Lower generalization error means better real-world performance.
Formula: $E_{train} = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i)$	Formula: $E_{gen} = \frac{1}{m} \sum_{j=1}^m L(y_j, \hat{y}_j)$
Example: A decision tree with very deep branches may predict training data almost perfectly ($E_{train} \approx 0$) but perform poorly on new data.	Example: A model that memorizes training data will have high E_{gen} because it fails to generalize to unseen patients/customers.



4. EXAMPLE OF PIPELINE (HEART DISEASE PREDICTION)



13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: (5 Marks) Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept EnjoySport. (a) Explain the initialization of S (specific boundary) and G (general boundary). (b) Show how S and G change with each training example. (c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.

13. Candidate-Elimination Algorithm

Dataset (Attributes: Sky, AirTemp, Humidity, Wind, Water, Forecast → Target Concept: EnjoySport)

Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
x_1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
x_2	Sunny	Warm	High	Strong	Warm	Same	Yes
x_3	Rainy	Cold	High	Strong	Warm	Change	No
x_4	Sunny	Warm	High	Strong	Cool	Change	Yes
x_5	Sunny	Cold	Normal	Strong	Cool	Change	No
x_6	Rainy	Warm	Normal	Strong	Warm	Same	Yes

(a) INITIALIZATION OF S AND G

1. Specific boundary (S):

Most specific hypothesis (contains only one hypothesis)
All attributes are set to the most specific value Φ (null symbol).

$$S_0 = \langle \Phi, \Phi, \Phi, \Phi, \Phi, \Phi, \Phi \rangle$$

2. General boundary (G):

Most general hypothesis (set of hypotheses)
Each attribute can take any value from its domain.

$$G_0 = \{ \langle ?, ?, ?, ?, ?, ?, ? \rangle \}$$

where ? represents "any value".

Legend

Φ : Most specific value (no value)

? : Any value

This ensures that

$$S_0 \subseteq C \subseteq G_0$$

where C is the set of all candidate hypotheses consistent with training examples seen so far.

(b) CHANGE OF S AND G WITH EACH TRAINING EXAMPLE

Step	Training Example (x_i)	Class (EnjoySport)	Specific Boundary (S)	General Boundary (G)
0 (Init.)	—	—	$\langle \Phi, \Phi, \Phi, \Phi, \Phi, \Phi, \Phi \rangle$	$\{ \langle ?, ?, ?, ?, ?, ?, ? \rangle \}$
1	$x_1 = \langle \text{Sunny, Warm, Normal, Strong, Warm, Same} \rangle$	Yes	$\langle \text{Sunny, Warm, Normal, Strong, Warm, Same} \rangle$	$\{ \langle ?, ?, ?, ?, ?, ?, ? \rangle \}$
2	$x_2 = \langle \text{Sunny, Warm, High, Strong, Warm, Same} \rangle$	Yes	$\langle \text{Sunny, Warm, ?, Strong, Warm, Same} \rangle$	$\{ \langle ?, ?, ?, ?, ?, ?, ? \rangle \}$
3	$x_3 = \langle \text{Rainy, Cold, High, Strong, Warm, Change} \rangle$	No	$\langle \text{Sunny, Warm, ?, Strong, Warm, Same} \rangle$	$\{ \langle \text{Sunny, ?, ?, ?, ?, ?} \rangle, \langle ?, \text{Warm, ?, ?, ?, ?} \rangle, \langle ?, ?, ?, ?, ?, \text{Same} \rangle \}$
4	$x_4 = \langle \text{Sunny, Warm, High, Strong, Cool, Change} \rangle$	Yes	$\langle \text{Sunny, Warm, ?, Strong, ?, ?} \rangle$	$\{ \langle \text{Sunny, ?, ?, ?, ?, ?} \rangle, \langle ?, \text{Warm, ?, ?, ?, ?} \rangle \}$
5	$x_5 = \langle \text{Sunny, Cold, Normal, Strong, Cool, Change} \rangle$	No	$\langle \text{Sunny, Warm, ?, Strong, ?, ?} \rangle$	$\{ \langle \text{Sunny, Warm, ?, ?, ?, ?} \rangle \}$
6	$x_6 = \langle \text{Rainy, Warm, Normal, Strong, Warm, Same} \rangle$	Yes	$\langle ?, \text{Warm, ?, Strong, ?, ?} \rangle$	$\{ \langle \text{Sunny, Warm, ?, ?, ?, ?} \rangle \}$

Final Version Space:

$$S_{\text{final}} = \langle ?, \text{Warm, ?, Strong, ?, ?} \rangle$$

$$G_{\text{final}} = \{ \langle \text{Sunny, Warm, ?, ?, ?, ?} \rangle \}$$

All hypotheses between S_{final} and G_{final} (inclusive) are consistent with all training examples.

(c) COMPARISON WITH FIND-S ALGORITHM

Aspect	Candidate-Elimination	Find-S
Approach	Maintains both a specific boundary (S) and a general boundary (G) to represent the version space.	Maintains only the most specific hypothesis that is consistent with all positive examples.
Hypotheses Maintained	Set of all hypotheses between S and G ($S \subseteq C \subseteq G$).	Single hypothesis h (most specific).
Use of Negative Examples	Uses both positive and negative examples to generalize S and specialize G.	Uses only positive examples; ignores negative examples.
Generalization Capability	More general. Considers boundaries of the concept, so can converge to a more accurate version space.	Less general. May produce overly specific hypothesis (overfitting).
Convergence Speed	Usually slower (maintains sets and updates both boundaries).	Faster (updates a single hypothesis).
Output	Version space (set of possible hypotheses).	Single hypothesis.
Risk of Overfitting	Lower (because negative examples restrict G).	Higher (cannot exclude unseen negative cases).



Key Takeaway:

Candidate-Elimination is more powerful than Find-S because it uses both positive and negative examples and returns the entire hypothesis space, leading to better generalization.

Notation:

Φ – most specific value (no value)

? – any value

S – specific boundary

G – general boundary

C – candidate hypothesis space

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)(a) Justify the choice of regression model (Linear Regression, Decision Tree, or RandomForest).(b) Explain feature scaling and handling missing data.(c) Describe how overfitting can be detected and avoided.

